Name: _____

TOPIC TEST 2

Chapter 5 Binomial distribution

- Time allowed: 70 minutes
- CAS calculators may be used.
- Part A: Five multiple-choice questions (5 marks).
- Part B: Six short-answer questions (37 marks).
- Part C: Two analysis questions (14 marks).

Part A

Multiple-choice questions

Five multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 In the wet season there is a 40% chance that there will be rainfall on any day. Rainfall on any day is independent of rainfall on any other day. What is the probability that there will be rainfall on four days in a week in the wet season?

A $(0.4)^4$

B $35 \times (0.4)^4 \times (0.6)^3$

C $7 \times 0.4 \times 0.6$

D $21 \times (0.4)^4 \times (0.6)^2$

E $\frac{4}{7}$

[1 mark]

2 X has a geometric distribution where X is the number of failures before the first success. If the probability of success is 0.7. Then $P(X = 4)$ is equal to:

A 0.0057

B 0.0187

C 0.1029

D 0.0723

E 0.0519

[1 mark]

3 The dial of a spinner has ten equally sized segments numbered 1, 2, 3, …, 10. The spinner is spun and the number on which the spinner comes to rest is noted. This is done a total of 8 times. If X is the number of even numbered segments on which the spinner rests, then which of these best describes the probability distribution for X?

A the geometric distribution with $p = 0.5$

B the binomial distribution with $n = 8$ and $p = \frac{8}{10}$

C the binomial distribution with $n = 10$ and $p = 0.5$

D the binomial distribution with $n = 8$ and $p = 0.5$

E the uniform distribution with $n = 10$ and $p = 0.5$

[1 mark]

Use the hypergeometric probability experiment defined by $P(X = 4) = \dfrac{\binom{15}{4}\binom{35}{6}}{\binom{50}{10}}$ to answer questions **4** and **5**.

4 How many successes are in the population for this experiment?

A 15

B 4

C 10

D 6

E 35

[1 mark]

5 What is the greatest value of X?

A 35

B 15

C 6

D 4

E 10

[1 mark]

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Part B

Short-answer questions

Six short-answer questions
31 marks
Show your working where appropriate.

1 A fair die is rolled 6 times. Find the probability of obtaining:

a a 5 on the third roll only

b a 5 on the first and second rolls only

c exactly three 5s.

[7 marks]

2 In a production line it is known that 18% of all finished products will have a defect. If X denotes the number of products that are defect free before the first one that has a defect, calculate:

a $P(X = 7)$

b $P(X \leq 7)$

c $E(X)$ and explain the result.

[9 marks]

3 A binomial variable, X, has the probability function:

$P(X = x) = \binom{7}{x}(0.28)^x(0.72)^{7-x}$.

Find:

a the number of trials

b the probability of success in any trial

c the probability distribution as a table.

[4 marks]

4 A box contains 40 computer chips of which 15% are known to be faulty. If a sample of 7 chips is selected at random and tested, calculate the probability that:

a two will be defective

b at least two will be defective.

[6 marks]

5 A binomial experiment has 10 trials. The probability of success in any trial is 0.4. If X = number of successes, calculate the probability that:

a $X = 3$

b X is at least 3

c X is no more than 3.

[7 marks]

6 Find the required probabilities in each of the following situations.

a A binomial experiment has 12 trials and $p = 0.3$. Find the probability that there are 7 successes obtained from the 12 trials.

b Find the binomial probability of 8 successes from 17 trials, where the probability of success is 0.7.

[4 marks]

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Part C

Analysis questions

Two analysis questions
14 marks
Show your working where appropriate.

1 A furniture manufacturer orders 400 three-metre lengths of timber from a timber mill. A length of timber is unacceptable if it contains knots. To determine whether or not to accept the order, the manufacturer randomly selects 10 lengths of timber and inspects them for knots. The manufacturer will only accept the full order if there are no defective lengths of timber in those that have been randomly selected for examination. What is the probability that none of the lengths of timber in the sample has knots if:

a 5% of the lengths of timber in the full order are known to have knots

b 15% of the lengths of timber in the full order are known to have knots.

[7 marks]

2 In the general population, it is known that 12% of all people carry a gene that predisposes them to a condition known as Syndrome K. Twenty-five people are randomly selected as a study group to determine if any are carriers of the gene.

a How many in the study group would you expect to be carriers of the gene?

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b What is the probability that the number of people in the study group who carry the gene is within one standard deviation of the mean?

[7 marks]

[Total marks: 56]

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Answers

Part A – Multiple-choice questions [1 mark each]

1 This is a Bernoulli trial with $n = 7$, $p = 0.4$ and $q = 0.6$.

Let X be the number of rainy days.

$$P(X = 4) = \binom{7}{4} \times (0.4)^4 \times (0.6)^2$$
$$= 35 \times (0.4)^4 \times (0.6)^2$$

$\therefore$ **B** **[1 mark]**

2 $p = 0.7$ and $q = 0.3$

$X = 4$

$$P(X = 4) = q^4 p$$
$$= (0.3)^4(0.7)$$
$$= 0.00567\ldots$$

$\therefore$ **A** **[1 mark]**

3 There are only two outcomes – an even number and an odd number.

$p = 0.5$ and $q = 0.5$

p and q are constant and independent.

There are 8 trials so $n = 8$

These are the conditions for a binomial experiment with $n = 8$ and $p = 0.5$.

$\therefore$ **D** **[1 mark]**

4 For this hypergeometric probability experiment $P(X = 4) = \dfrac{\binom{15}{4}\binom{35}{6}}{\binom{50}{10}}$

For the general hypergeometric probability $P(X = x) = \dfrac{\binom{k}{x}\binom{N-k}{n-x}}{\binom{N}{n}}$

By comparison, $k = 15$, $N = 50$, $n = 10$ and $x = 4$.

Number of successes in population $= k = 15$.

$\therefore$ **A** **[1 mark]**

5 $k = 15$, $N = 50$, $n = 10$ and $x = 4$

There are only 10 trials, so the greatest value that X can be is 10.

$\therefore$ **E** **[1 mark]**

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Part B – Short-answer questions

1 a Success (S) = rolling a 5

Failure (F) = rolling a number other than 5

$p = \frac{1}{6}$ and $q = \frac{5}{6}$ **[1 mark]**

$P(5 \text{ on the third roll only}) = P(FFSFFF)$ **[1 mark]**

$= \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6}$

≈ 0.0670 **[1 mark]**

b $P(6 \text{ on 1st and 2nd rolls only}) = P(SSFFFF)$ **[1 mark]**

$= \frac{1}{6} \times \frac{1}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6}$

≈ 0.0134 **[1 mark]**

c $P(X = 3) = \binom{6}{3} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^3$ **[1 mark]**

≈ 0.0536 **[1 mark]**

2 a $p = 18\% = 0.18$

$q = 1 - 0.18 = 0.82$ **[1 mark]**

$P(X = x) = (0.82)^x(0.18)$ **[1 mark]**

Let $X = 7$,

$P(X = 7) = (0.82)^7(0.18)$

≈ 0.0449 **[1 mark]**

b $P(X \leq 7) = P(X = 0) + P(X = 1) + P(X = 2) + \ldots + P(X = 7)$

$= (0.18) + (0.82)^1(0.18) + (0.82)^2(0.18) + \ldots + (0.82)^7(0.18)$ **[1 mark]**

This is a geometric series: $a + ar + ar^2 + \ldots + ar^{n-1}$.

$a = 0.18$, $r = 0.82$ and $n = 8$. **[1 mark]**

$$S_n = \frac{a(1 - r^8)}{1 - r}$$

$$= \frac{0.18\left[1 - (0.82)^8\right]}{1 - 0.82}$$ **[1 mark]**

$$= 1 - (0.82)^8$$

$$\approx 0.7956$$ **[1 mark]**

c $E(X) = \frac{1 - p}{p}$ **[1 mark]**

$$= \frac{1 - 0.18}{0.18}$$

$$\approx 4.56$$

$E(X) \approx 4.56$, so you can expect that there will be 5 without a defect before the first defective product occurs. **[1 mark]**

3 a For any binomial experiment, $P(X = x) = \binom{n}{x} p^x q^{n-x}$

For this binomial experiment, $P(X = x) = \binom{7}{x} (0.28)^x (0.72)^{7-x}$

$n = 7$ **[1 mark]**

b $p = 0.28$ **[1 mark]**

c $P(X = 0) = \binom{7}{0} (0.28)^0 (0.72)^7 \approx 0.1003$

$P(X = 1) = \binom{7}{1} (0.28)^1 (0.72)^6 \approx 0.2731$

$P(X = 2) = \binom{7}{2} (0.28)^2 (0.72)^5 \approx 0.3186$

And so on. **[1 mark]**

x	0	1	2	3	4	5	6	7
$p(x)$	0.1003	0.2731	0.3186	0.2065	0.0803	0.0187	0.0024	0.0001

[1 mark]

4 a Let X = number of faulty chips

$N = 40$

$n = 7$

$k = 0.15 \times 40 = 6$ **[1 mark]**

$$P(X = x) = \frac{\binom{k}{x}\binom{N-k}{n-x}}{\binom{N}{n}}$$

$$P(X = 2) = \frac{\binom{6}{2}\binom{40-6}{7-2}}{\binom{40}{7}}$$ **[1 mark]**

$$\approx \frac{\binom{6}{2}\binom{34}{5}}{\binom{40}{7}}$$

≈ 0.2239 **[1 mark]**

b $P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$ **[1 mark]**

$$P(X = 0) = \frac{\binom{6}{0}\binom{34}{7}}{\binom{40}{7}} \approx 0.2886$$

$$P(X = 1) = \frac{\binom{6}{1}\binom{34}{6}}{\binom{40}{7}} \approx 0.4328$$

$P(X = 2) \approx 0.2239$ **[1 mark]**

$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$

$\approx 00.2886 + 0.4328 + 0.2239$

≈ 0.9453 **[1 mark]**

5 a This is a binomial experiment.

$n = 10$

$p = 0.4$

$q = 0.6$ **[1 mark]**

$$P(X = x) = \binom{n}{x} p^x q^{n-x}$$

$$P(X = 3) = \binom{10}{3}(0.4)^3(0.6)^7$$ **[1 mark]**

$= 0.214\,991\ldots$

≈ 0.2150 **[1 mark]**

b $P(X \text{ is at least } 3) = P(X \geq 3)$

$= 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$ **[1 mark]**

$$P(X = 0) = \binom{10}{0}(0.4)^0(0.6)^{10} = 0.006\,047\ldots$$

$$P(X = 1) = \binom{10}{1}(0.4)^1(0.6)^9 = 0.040\,311\ldots$$

$$P(X = 2) = \binom{10}{2}(0.4)^2(0.6)^8 = 0.120\,932\ldots$$

$P(X \geq 3) = 1 - (0.006\,047\ldots + 0.040\,311\ldots + 0.120\,932\ldots)$

≈ 0.8327 **[1 mark]**

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c $P(X \text{ is no more than } 3) = P(X \leq 3)$ **[1 mark]**

$= P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$

$= 0.006\,047\ldots + 0.040\,311\ldots + 0.120\,932\ldots + 0.214\,991\ldots$

≈ 0.3823 **[1 mark]**

6 a This is a binomial experiment.

$p = 0.3$

$n = 12$

$x = 7$ **[1 mark]**

Intersection of row where $x = 7$ (from the left) and column where $p = 0.3$ (from above)

$= 0.0291$

$P(X = 4) \approx 0.0291$ **[1 mark]**

b This is a binomial experiment.

$p = 0.8$

$n = 17$

$x = 8$ **[1 mark]**

Intersection of row where $x = 8$ (from the right) and column where $p = 0.8$ (from below)

$= 0.0021$

$P(X = 7) \approx 0.0021$ **[1 mark]**

Part C – Analysis questions

1 a Success = a length of timber with knots

$N = 400$

$n = 10$

$k = 0.05 \times 400 = 20$ **[1 mark]**

$$P(X = 0) = \frac{\binom{20}{0}\binom{400 - 20}{10 - 0}}{\binom{400}{10}}$$

$$= \frac{\binom{20}{0}\binom{380}{10}}{\binom{400}{10}}$$ **[1 mark]**

$= 0.595\,144\ldots$

When 5% of all lengths of timber have knots, the probability that none of the sample lengths of timber has knots is about 60%. **[1 mark]**

b $N = 400$

$n = 10$

$k = 0.15 \times 400 = 60$ **[1 mark]**

$$P(X = 0) = \frac{\binom{60}{0}\binom{400-60}{10-0}}{\binom{400}{10}}$$

$$= \frac{\binom{60}{0}\binom{340}{10}}{\binom{400}{10}}$$ **[1 mark]**

$= 0.192\,937\ldots$

When 15% of all lengths of timber have knots, the probability that none of the sample lengths of timber has knots is about 19%. **[1 mark]**

2 a This is a binomial experiment.

$n = 25$

$p = 0.12$ and $q = 0.88$ **[1 mark]**

$E(X) = np$ **[1 mark]**

$= 25 \times 0.12$

$= 3$

You could expect that about three people in the study group will be carriers of the gene. **[1 mark]**

b $\text{SD}(X) = \sigma = \sqrt{npq}$

$= \sqrt{25 \times 0.12 \times 0.88}$

≈ 1.62 **[1 mark]**

$\mu + \sigma \approx 3 + 1.62 = 4.62$

$\mu - \sigma \approx 3 - 1.62 = 1.38$ **[1 mark]**

$P(\mu - \sigma \leq X \leq \mu + \sigma) = P(1.38 \leq X \leq 4.62)$

$= P(2 \leq X \leq 4)$ **[1 mark]**

$= P(X = 2) + P(X = 3) + P(X = 4)$

$= 0.2283\ldots + 0.2387\ldots + 0.1790\ldots$

≈ 0.6461

There is about a 65% chance that from 2 to 4 people in the study group will be carriers of the gene. **[1 mark]**

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